

Translating Noise into Strategic Insight

Project 2: Exploratory Data Analysis & Data Forensics

Umaina Asif Khan

Batch 2026 | Digital Attribution Report

| Executive Summary

Key Diagnostics

- **Standardized** Coupon Code taxonomy across 1,200 transactions.
- **Mitigated** 25% data gap in revenue attribution.
- **Identified** WINTER15 as the primary revenue driver.

"Data forensics is not just about cleaning; it is about uncovering the pulse of the business hidden behind inconsistent entries."

| Strategic Objectives



Forensic Cleaning

Removing duplicates and inconsistencies to establish a 'Single Source of Truth'.



Revenue Attribution

Mapping Coupon performance to understand customer behavior and conversion.



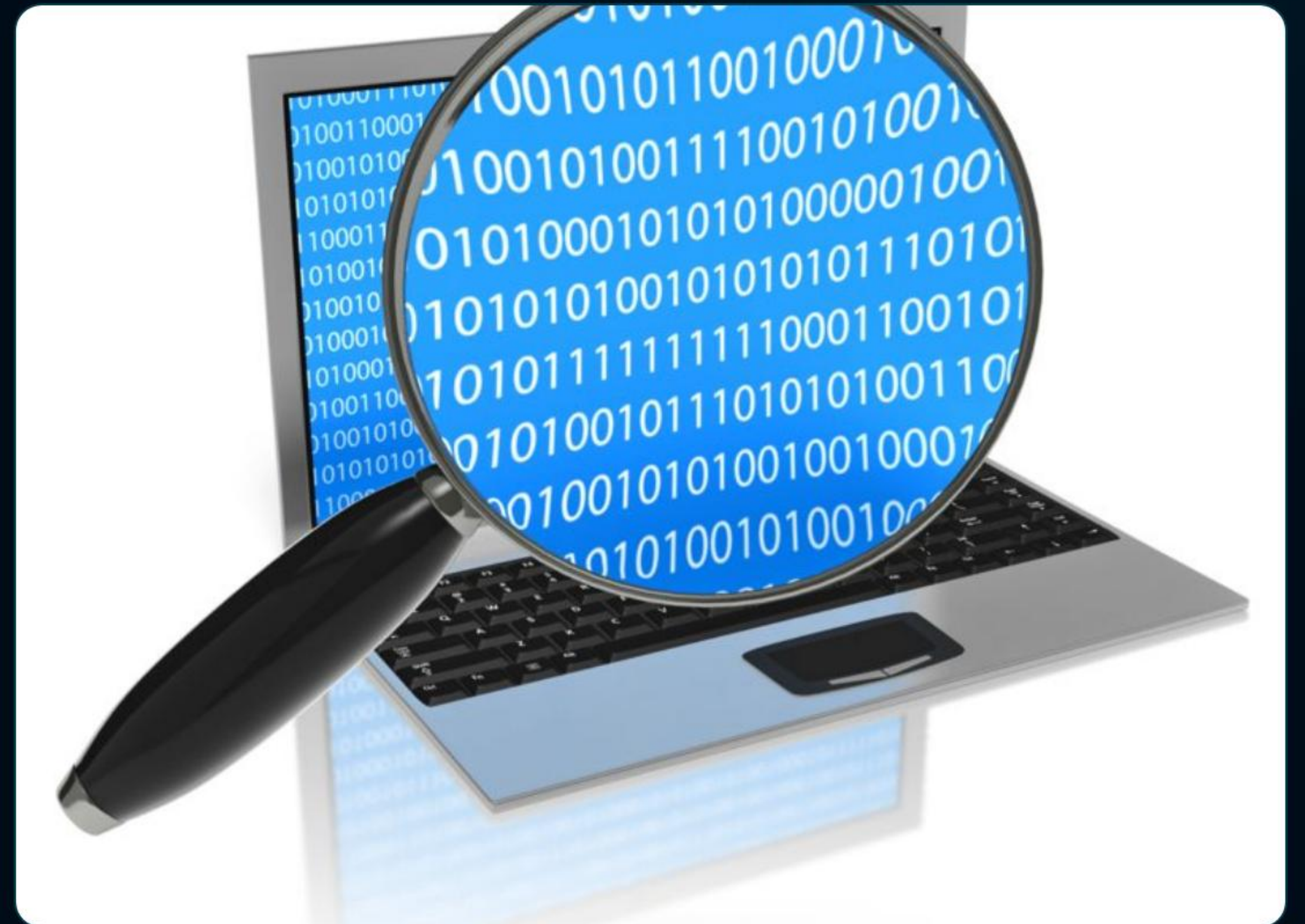
Data Integrity

Closing the gaps in checkout funnel tracking and identifying system leakages.

| Forensic Methodology

Our approach utilized **Python-based automation** to perform deep-tissue data cleaning:

- ✓ **Deduplication:** Identifying redundant records.
- ✓ **Standardization:** Normalizing code formats.
- ✓ **Imputation:** Smart-filling missing gaps.



| Standardizing Coupon Codes

Before (Noise)

"Save 10", "save 10", "SAVE 10", "nan"

Fragmented data makes comparison impossible.

After (Signal)

"SAVE10", "SAVE10", "SAVE10", "Unknown"

Standardized format allows for accurate Pivot analysis.

Using `str.strip()` and `str.upper()` we achieved 100% uniformity.

| Outlier Mitigation (IQR)

1.5x

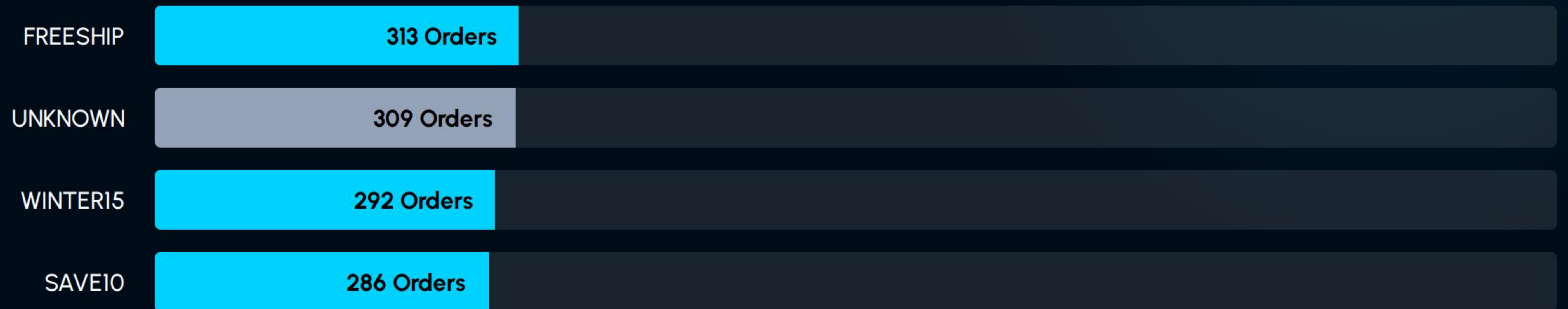
IQR Threshold

Refining the Median

By applying the **Interquartile Range (IQR)** method, we successfully clipped extreme values in 'UnitPrice' and 'Quantity'.

This ensures our average sales figures reflect **typical customer behavior** rather than data noise or anomalies.

Coupon Performance Analysis



WINTER15 generated the highest yield per transaction despite lower frequency.

| The Attribution Gap

25% Missing Footprint

309 transactions failed to log a specific Coupon Code at checkout. This "Unknown" segment represents a significant risk to marketing budget allocation.

Without accurate tracking, it is impossible to calculate true **ROAS (Return on Ad Spend)** for non-coupon campaigns.

600 × 400

Data Integrity Comparison

Metric	Raw Dataset	Cleaned (Forensic)	Impact
Duplicates	Present	Zero	Accurate Counts
Missing Coupons	NaN / Blanks	Labled "Unknown"	Segmentable Data
Outliers	Extreme Variance	IQR Clipped	Stable Averages
Naming	Mixed Case/Spaces	UPPER_SNAKE	Standardized Pivot

| Strategic Recommendations



Technical Audit

Review checkout API calls to ensure CouponCode is logged for every conversion event.



Funnel Design

Make the coupon field more prominent to encourage logging of intent-based identifiers.



WINTER15 Scale

Re-invest marketing budget into the WINTER15 campaign style as it shows the highest value.

"In God we trust. All others must bring data."

— W. Edwards Deming

By fixing the foundations of our dataset, we have empowered Umaima Asif Khan's analysis to drive real-world business results.

Thank You!

Umaina Asif Khan | Data Analytics Project Report

Questions?

Contact: Umaina Asif Khan | LMS Portal

| Image Sources



https://content.presentermedia.com/files/clipart/00011000/11672/computer_magnifying_glass_800_wht.jpg

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